

MYS BILLINGS, Russian Federation

WMO: 250620

Lat: 69.877N

Lon: 175.755E

Elev: 10

StdP: 14.69

Time Zone: 12.00 (E12)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.5	-35.3	-44.5	0.4	-38.1	-41.1	0.5	-35.1	34.2	8.4	30.1	-4.5	6.1	230	1.087

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	60.4	54.1	54.9	50.4	49.6	46.6	54.2	60.3	50.4	55.0	46.6	49.7	12.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.8	55.1	57.5	46.7	47.1	53.5	43.6	41.8	49.5	22.8	60.4	20.5	55.1	18.5	49.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.5	27.9	25.3	DB	-44.3	70.9	5.1	5.8	-47.9	75.1	-50.9	78.5	-53.8	81.8	-57.5	86.0	(4)
(5)			WB	-42.9	61.8	4.9	6.1	-46.5	66.2	-49.4	69.7	-52.1	73.1	-55.7	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	-12.1	-11.7	-10.2	4.0	22.4	35.7	40.2	39.1	35.6	23.6	8.9	-4.3
	DBStd	22.60	12.57	14.05	13.26	11.68	8.31	5.67	5.79	5.53	5.54	9.38	12.38	15.34
	HDD50	13014	1924	1728	1866	1379	855	432	315	343	436	819	1233	1684
	HDD65	18465	2389	2148	2331	1829	1320	878	769	802	883	1284	1683	2149
	CDD50	25	0	0	0	0	0	4	12	6	3	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	8	0	0	0	0	0	0	8	0	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	11.4	10.9	10.4	10.6	10.6	10.7	12.1	12.0	12.8	12.3	11.9	11.9	11.0
Precipitation	PrecAvg	17.9	1.3	1.4	0.9	0.9	1.1	1.1	1.9	2.0	2.0	2.2	1.8	1.3
	PrecMax	25.8	3.7	3.4	2.2	1.7	2.3	2.1	3.8	3.7	5.2	7.0	3.6	3.1
	PrecMin	11.9	0.2	0.1	0.0	0.4	0.2	0.5	0.5	0.7	0.5	0.5	0.4	0.2
	PrecStd	3.3	0.7	0.8	0.6	0.4	0.5	0.5	0.8	0.8	1.0	1.2	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	30.8	32.9	34.8	44.2	64.4	72.0	63.7	59.5	41.3	33.5	30.8
		MCWB	29.7	28.7	30.8	32.6	41.3	54.3	62.0	56.4	52.6	39.0	31.4	28.8
	2%	DB	23.4	19.9	26.2	27.9	37.2	57.2	62.5	57.1	52.8	37.4	31.5	25.1
		MCWB	22.2	18.8	24.7	26.6	34.8	50.2	56.5	52.0	48.9	36.5	30.4	23.9
	5%	DB	12.4	13.3	15.9	24.4	34.4	49.8	54.5	51.2	46.5	35.7	29.0	20.0
		MCWB	11.6	12.7	15.0	23.6	33.0	46.1	50.7	48.0	44.2	34.9	28.1	19.1
	10%	DB	3.5	6.4	9.0	20.9	32.7	44.0	48.5	46.4	41.1	34.1	25.7	16.4
		MCWB	2.9	5.9	8.3	20.1	31.7	41.3	45.8	44.2	39.4	33.1	24.6	15.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	28.6	31.1	32.6	41.1	54.6	63.5	57.7	53.0	39.3	31.8	28.8
		MCDB	31.6	30.8	32.9	34.7	44.1	64.4	69.7	62.4	58.7	41.0	32.4	30.4
	2%	WB	22.6	18.7	24.6	26.8	35.0	50.5	56.0	52.4	48.7	36.6	30.3	24.0
		MCDB	23.8	19.7	26.1	27.9	36.5	57.0	61.9	56.9	52.4	37.2	31.3	25.0
	5%	WB	11.6	12.6	15.1	23.6	33.1	46.2	51.0	48.1	43.7	35.1	28.0	19.1
		MCDB	12.2	13.2	16.0	24.4	34.1	50.3	54.8	51.4	46.5	35.7	28.9	19.9
	10%	WB	3.0	5.9	8.2	20.3	31.8	41.4	45.8	44.3	39.7	33.4	24.6	15.5
		MCDB	3.4	6.3	8.9	21.0	32.6	43.9	48.3	46.3	40.8	34.2	25.5	16.4
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.9	11.5	12.6	8.9	7.7	9.0	6.6	5.2	7.0	10.9	10.3
		MCDBR	19.8	17.4	17.3	12.7	8.9	18.9	21.8	17.7	13.9	4.9	9.5	13.9
		MCWBR	19.0	16.8	16.5	12.1	7.7	14.1	17.3	13.9	10.7	4.4	9.4	13.5
	5% WB	MCDBR	19.6	16.9	17.4	12.8	8.1	19.2	21.8	17.6	13.3	4.7	9.4	13.7
		MCWBR	19.0	16.3	16.6	12.2	7.1	14.3	17.6	14.1	10.4	4.2	9.3	13.4
Clear-Sky Solar Irradiance	taub	N/A	0.208	0.281	0.327	0.354	0.352	0.373	0.377	0.312	0.248	N/A	N/A	
	taud	N/A	2.025	2.032	1.960	2.012	2.232	2.311	2.313	2.449	2.121	N/A	N/A	
	Ebn at Noon	0	170	223	248	257	265	254	235	225	144	0	0	
	Edh at Noon	0	13	27	40	43	36	33	29	19	11	0	0	
All-Sky Solar Radiation	RadAvg	0	106	629	1370	1891	1938	1577	1007	516	154	0	0	
	RadStd	0	14	41	64	107	129	141	99	42	20	0	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page